

Lab10: Pirelli experience at Vizzola test track

12/12/2025

Abstract

This report analyzes a driving experience at the Pirelli Vizzola track using an RWD vehicle to validate lateral dynamics theories. Through steering pad and power-on/power-off maneuvers, the study examined stability limits and the impact of combined slip. The results highlighted the practical challenges of controlling unstable equilibrium and demonstrated how the interaction between longitudinal load transfer and tire friction limits fundamentally alters RWD handling stability.

Contents

1. Introduction.....3

2. Steering pad constant radius4

3. Power-on and power-off maneuvers6

 3.1 Power-on maneuver6

 3.2 Power-off maneuver7

4. Conclusion.....8

1. Introduction

The goal of this experience was to get familiar with the maneuvers that are usually simulated and used to have a better understanding of the lateral behavior of a vehicle. The experience was conducted with a RWD vehicle and three different scenarios were tested:

- 1) steering pad constant radius maneuver (Section 2)
- 2) power-on maneuver (Section 3.1)
- 3) power-off maneuver (Section 3.2)

2. Steering pad constant radius

A circular path with fixed radius R is run at increasing speed V , the steering angle δ is adjusted to follow the trajectory. Note that the handling diagram is usually derived under the assumption of steady state conditions; however, if the speed is increased slowly, it can be seen as a sequence of steady state conditions.

The aim of this part of the experience was to understand the difference between the two different working points that can be identified on the handling diagram. **From the theoretical point of view**, with reference to Figure 1, the working points are defined as the intersections between the handling diagram (continuous black line) and the dashed line that has equation in the $(\alpha_F - \alpha_r, \frac{a_y}{g})$ plane

$$\delta - \frac{lg}{V^2} * \frac{a_y}{g}$$

- $\delta \rightarrow$ steering angle, it is the shift of the line
- $V \rightarrow$ speed, it influences the slope of the line
- $l \rightarrow$ wheelbase
- $a_y \rightarrow$ lateral acceleration

Once the steering angle and the speed are defined, the working points are identified:

- point **A**: stable and in understeering condition
- point **B**: unstable and in oversteering condition

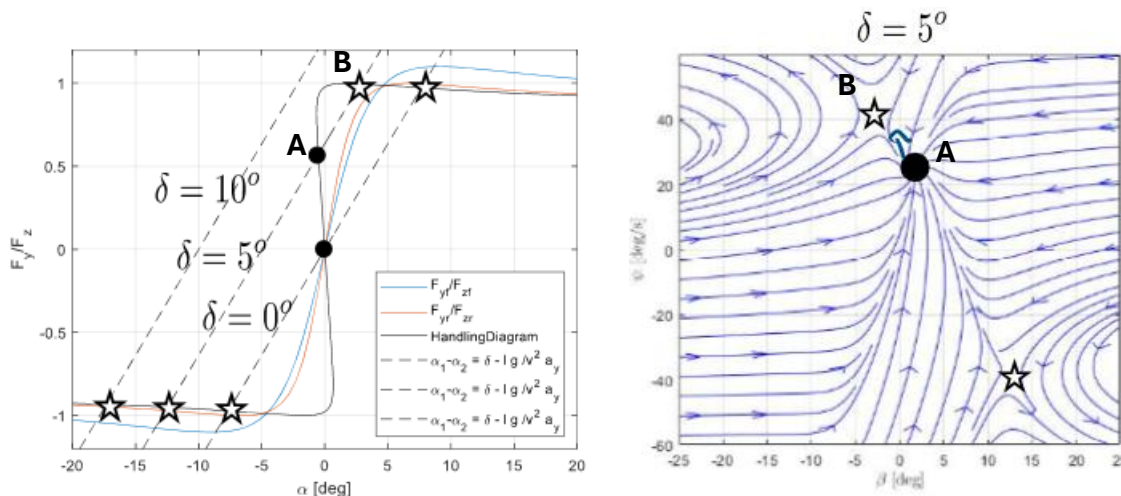


Figure 1

In reality, it is very hard to reach these two working conditions with the same velocity and steering angle, especially point B, that is associated with an unstable condition in which it is hard to control the vehicle.

For instance, the Pirelli driver was able to easily find a stable working point with the car in understeering condition while it was much harder to keep the car in an unstable, oversteering working point (a road vehicle was used, it was not meant to be driven in that way).

3. Power-on and power-off maneuvers

The aim of this part of the experience was to understand how is the load transferred from front and rear axle and from left to right during power on and power off maneuvers. Note that Figure 2 and Figure 3 are intended as a **qualitative description** of the maneuvers performed by the Pirelli test driver, the speed, steering angle that can be read are not the same.

3.1 Power-on maneuver

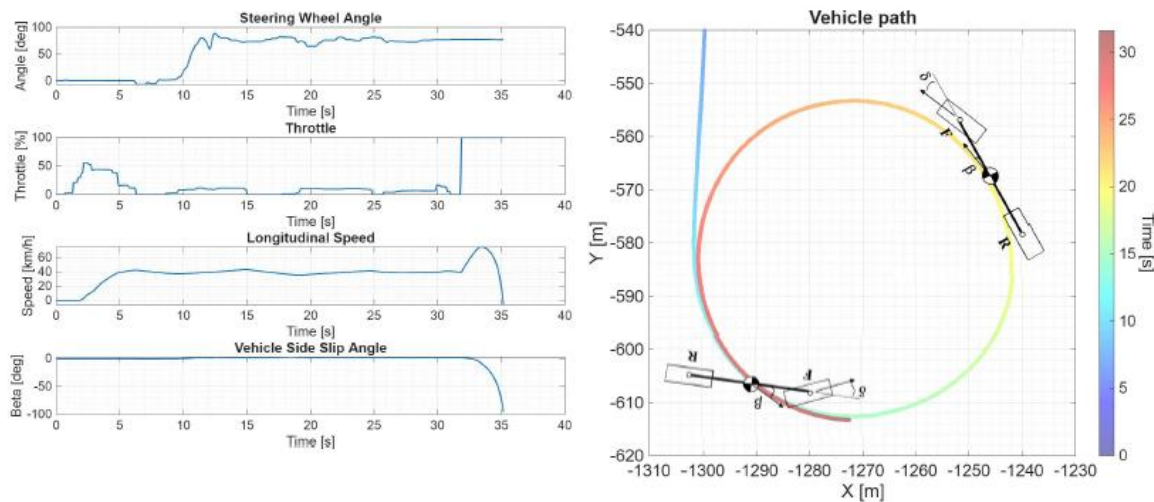


Figure 2

- a) Car is turning left $\rightarrow$ lateral load transfer $\rightarrow$ higher vertical load on the outer tires throughout the whole maneuver (higher capability of generating a lateral force)
- b) Power on $\rightarrow$ longitudinal load transfer $\rightarrow$ higher vertical load on the rear tires when the car accelerates (higher capability of generating a lateral force)
- c) RWD $\rightarrow$ the total grip that the rear tires can generate is shared between longitudinal and lateral. When the driver accelerates, the capability of the rear tires to generate lateral force decreases

Note that factor c is in contrast with factors $a, b \rightarrow$ the outcome of the maneuver depends on the interaction between the three factors. For example, if the rear lateral grip saturates before the front (factor c prevails), the rear axle begins to slide and the car behavior will be oversteering. On the other hand, if the front grip saturates before the rear one, the car behavior will be understeering.

In the real test experienced the rear of the vehicle began to slide (oversteering behavior).

3.2 Power-off maneuver

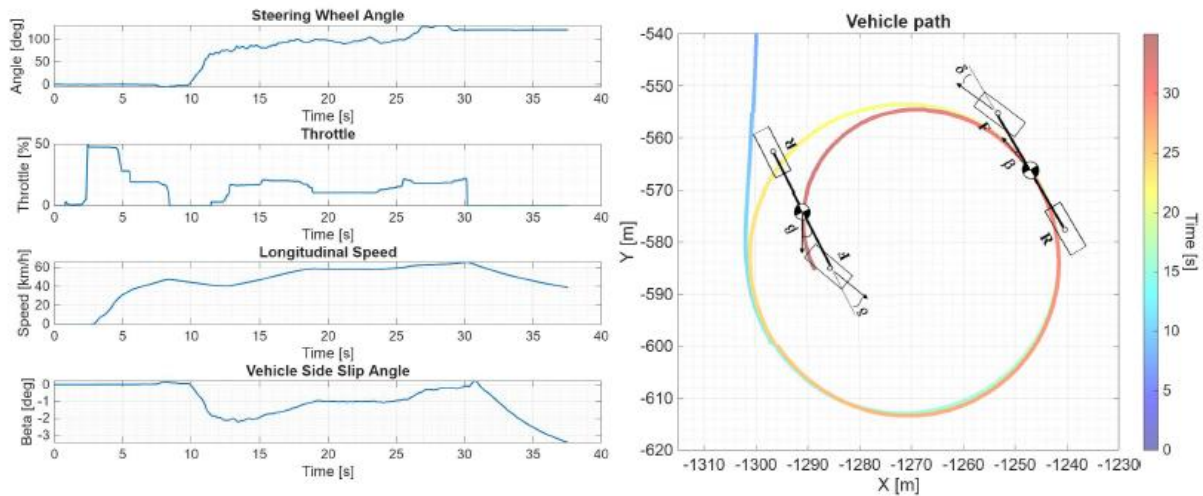


Figure 3

- a) Car is turning left $\rightarrow$ lateral load transfer $\rightarrow$ higher vertical load on the outer tires throughout the whole maneuver (higher capability of generating a lateral force)
- b) Power off $\rightarrow$ longitudinal load transfer $\rightarrow$ higher vertical load on the front tires when the throttle is released (higher capability of generating a lateral force)
- c) RWD $\rightarrow$ the total grip that the rear tires can generate is shared between longitudinal and lateral. When the driver lifts off the throttle, the rear tires are still required to generate a longitudinal braking force, which in any case is lower than the longitudinal force required during a power on maneuver.

Note that factor once again the outcome of the maneuver depends on the interaction between the three factors.

In the real test experienced the radius of the trajectory decreased, however, it was hard to feel if the car behavior was understeering or oversteering (even in the simulation presented the day before the car behavior is almost neutral).

4. Conclusion

The experience validated theoretical models against real-world RWD behavior. Steering pad tests confirmed that while stable understeer is easily managed, the unstable oversteer limit is practically difficult for a driver to maintain. Power-on maneuvers illustrated the friction circle conflict, where traction demand overwhelmed the rear axle's lateral capacity despite increased vertical load, leading to oversteer. Conversely, power-off maneuvers demonstrated a "tuck-in" effect, where forward load transfer increased front grip and tightened the trajectory without causing severe instability. Ultimately, the tests proved that vehicle handling is dynamically determined by the real-time interaction between simultaneous load transfers and tire saturation limits.